

MEETANSH KHARBANDA

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EDUCATION

BASc - Bachelors of Applied Science- Computer Engineering

Queen's University

Sep 2021 – Apr 2025

Kingston, ON

- **Relevant Coursework:** Object Oriented Programming (Java), Data Structures & Algorithms (C, Python), Comp. Vision with Deep Learning, Data Analytics, Operating Systems (C, Linux), Artificial Intelligence, ML & Deep Learning

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Java, SQL, HTML, CSS, PHP

Technologies & Tools: Github, VS Code, Solidworks, Docker, Azure, JIRA, Figma

Libraries/ Frameworks: OpenCV, TensorFlow, pandas, NumPy, PyTorch, Scikit-learn, Hugging Face, Selenium

DBMS & Operating Systems: MySQL, SQLite, Windows, Linux

EXPERIENCE

AI Consultant

QMIND

Sep 2022 – Apr 2024

Kingston, ON

- Worked in a team of 4 to develop an **NLP** chatbot in Python for a leading financial solutions company.
- Employed Selenium to scrape **90** articles as data from the client's website and fed it to the transformer.
- Integrated **Hugging Face** transformer library to assess text similarity, resulting in an accuracy rate of **98%**.
- More than **100,000** users can now receive accurate and precise responses to their queries.

PROJECTS

Capstone- Oil Spill Detection | OpenCV, PyTorch, Google Cloud

Present

- Developed a deep learning model to detect oil spills in oceans using Sentinel-1 Synthetic Aperture Radar (SAR) data, aiming to enhance environmental monitoring capabilities.
- Implemented a **UNet-MobileNetV3** based architecture for oil spill segmentation. Actively refining model accuracy, aiming to exceed **90%**. Currently, achieved a **Mean IoU of 0.74**, pixel accuracy: **94.92%**
- Leveraging cloud computing platforms, including Google Earth Engine, to process large datasets in real-time, ensuring timely identification and mitigation of marine pollution.

Ensemble Image Classification | Python, PyTorch, CIFAR-100

Oct 2024 – Nov 2024

- Trained and evaluated AlexNet, VGG16, and ResNet18 models on the CIFAR-100 dataset, achieving an average accuracy of **78.63%**, **82.58%**, and **86.05%**, respectively.
- Designed and implemented ensemble methods (maximum probability, probability averaging, majority voting) to improve classification performance.
- Achieved consistent improvements in top-1 and top-5 classification performance through ensemble methods, demonstrating a **10%** reduction in error rates for fully converged models.

Length of Stay Prediction | Python, Jupyter Notebooks, PyTorch, Numpy

Jan 2024 – Apr 2024

- Cleaned and pre-processed a large dataset of **4,55,500** patient records, addressing missing values and outliers, resulting in a **20%** improvement in data quality.
- Utilized 5-fold cross-validation to ensure robust model performance, achieving an average F1 score of **72%**.
- Collaborated with healthcare domain experts to incorporate domain-specific knowledge, resulting in a **7%** improvement in model accuracy on clinically relevant predictions.

Fraud Analysis | KNIME, Python, Colab

Sep 2023 – Dec 2023

- Conducted a comprehensive exploration of a numeric fraud dataset to understand the characteristics and patterns of fraudulent activities within the company.
- Applied clustering methods like **k-means**, **hierarchical**, and **DB Scan** to reveal distinct groups in fraudulent activities, gaining insights into various fraud patterns and boosting the model's predictive power by **10%**.
- Integrated Decision Trees to identify potential fraudulent cases, achieving an accuracy of **92%** on the test set.